



Figure 1: iLO home page

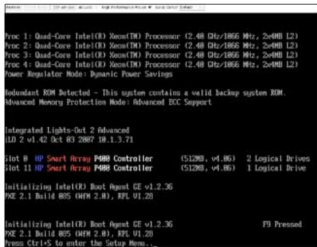


Figure 2: Virtual screen remotely displaying server console

a rational point of view, this would be a huge loss of time. Maybe I ought to try and restore GRUB first—but how? Fortunately, Hewlett-Packard hardware comes with the iLO service on-board [1].

iLO—integrated Lights-Out

For years, OEMs of complex computer systems have tried to expand the functionality of their enterprise-level product lines, for example, by utilising out-of-band management (a system feature using which working traffic goes through a casual network card, while service traffic comes via a dedicated channel). In fact, a server has an additional Ethernet controller that's intended for service functions only. For instance, through this dedicated channel, access to the server is organised—it can be seen as a secured and absolutely independent connection to a server. Don't misuse it with a VLAN subnet, because it's a completely separate network. Thus far, the traffic that goes through this dedicated channel is truly for management purposes. So, you can establish a virtual TTY connection to a UNIX server, or organise a fully featured graphical access to the server by means of iLO—the integrated Lights-Out service—and not be afraid of overloading other channels.

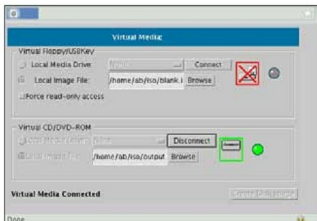


Figure 3: Virtual media is connected in a blink of an eye

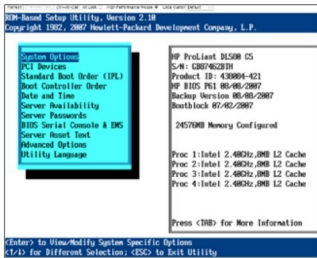


Figure 4: HP ROM-based Setup Utility (virtual screen)

Basically, iLO is a special portion of firmware (i.e., BIOS) developed by Hewlett-Packard that transforms TCP/IP packets into serial line signals. There are competitors to iLO implemented by other vendors. In particular, Sun Microsystems had developed a feature pack called Advanced Lights Out Management (ALOM). Fujitsu has Integrated Remote Management Controller (iRMC), and Dell has Dell Remote Access Controller, or DRAC. All of them have one goal—to offer a virtual KVM.

The HP iLO implementation is available in two options: you can either connect to the server via SSH and perform all tasks in a text-console session, or you can launch your Web browser and have a full-featured graphical desktop environment in order to manipulate the hardware. In the latter case, you get a far more functional solution, because you can use the local screen and peripherals by your side as virtual devices (the real ones are left in the server room). We'll be using only the advanced features of iLO. Let's get started with Mozilla Firefox, and locate the URL to the server's iLO home page. You need to have a Web browser with a Java plug-in to log in to iLO.